1️⃣ np.concatenate()

✅ Definition

Joins arrays **along an existing axis**.

🔹 Syntax

np.concatenate((arr1, arr2), axis=0)

🔹 Example

```
import numpy as np


a = np.array([[1,2],[3,4]])

b = np.array([[5,6],[7,8]])


# Join row-wise (default axis=0)

np.concatenate((a,b), axis=0)
```

Result:

```
[[1 2]

 [3 4]

 [5 6]

 [7 8]]
```

```
# Join column-wise

np.concatenate((a,b), axis=1)
```

Result:

```
[[1 2 5 6]

 [3 4 7 8]]
```

⚠ Important

- Arrays must have **same shape except in the joining axis**

🔹 2️⃣ np.vstack() (Vertical Stack)

✅ **Definition**

Stacks arrays **row-wise** (vertical stacking).
Equivalent to axis=0 concatenate.

🔹 **Example**

np.vstack((a, b))

Same as:

np.concatenate((a,b), axis=0)

🔹 3️⃣ **np.hstack() (Horizontal Stack)**

✅ **Definition**

Stacks arrays **column-wise** (side by side).
Equivalent to axis=1 concatenate (for 2D).

🔹 **Example**

np.hstack((a, b))

Same as:

np.concatenate((a,b), axis=1)

🔹 4️⃣ **np.column_stack()**

✅ **Definition**

Stacks 1D arrays as columns into a 2D array.

🔹 **Example**

a = np.array([1,2,3])

b = np.array([4,5,6])

np.column_stack((a,b))

Result:

[[1 4]

[2 5]

[3 6]]

🔥 **Important Difference**

- hstack() with 1D arrays → gives 1D result
- column_stack() → converts to 2D columns

🔹 5️⃣ **np.split()**

✅ **Definition**

Splits array into **equal parts**

🔹 **Syntax**

np.split(array, sections, axis=0)

🔹 **Example**

arr = np.arange(8)

np.split(arr, 4)

Result:

[array([0,1]),

array([2,3]),

array([4,5]),

array([6,7])]

⚠️ **Important**

- Must divide **equally**
- Otherwise → Error

🔹 6️⃣ **np.array_split()**

✅ **Definition**

Splits array into **unequal parts allowed**

🔹 **Example**

arr = np.arange(7)

np.array_split(arr, 3)

Result:

[array([0,1,2]),

array([3,4]),

array([5,6])]

🔥 **Difference from split**

split()	**array_split()**
Equal parts only	Unequal allowed
Gives error if not divisible	No error

🔹 **Splitting with axis (2D example)**

a = np.arange(12).reshape(3,4)

np.split(a, 2, axis=1)

Splits columns into 2 equal parts.

🔥 **Summary Table**

Function	**Purpose**
concatenate	General joining
vstack	Stack rows
hstack	Stack columns
column_stack	Make columns from 1D
split	Equal split
array_split	Flexible split